Ordinal Objectives and Metric Disagreement in Fine-Grained Text Sentiment Classification

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Abstract

Fine-grained sentiment labels are ordered, but standard cross-entropy (CE) does not distinguish an adjacent error from a polarity reversal. We compare CE, four ordinal soft-label objectives, CORN, a cumulative link model (CLM), and a kappa-tuned CLM on movie reviews, Yelp reviews, and financial sentiment. Results are descriptive means and standard deviations over 20 predetermined seeds. Across the 12 soft-label-versus-CE comparisons, observed mean quadratic weighted kappa (QWK) is higher in eight, tied in one, and lower in three. Improvements transfer less often to exact-match Accuracy or mean absolute error (MAE), and leading objectives vary by metric and dataset. In the archived seed-42 diagnostics, higher QWK can coincide with a larger share of errors spanning two or more classes. The CLM-family heads have lower observed aggregate performance; in the same diagnostics, they also concentrate predictions at scale extremes. A finance-domain, sentiment-adapted checkpoint yields higher observed QWK than BERT for all eight objectives in one conditional comparison, but this comparison does not isolate domain pretraining or establish a general encoder effect. Ordinal supervision therefore depends on the evaluation metric, dataset, and configuration rather than being a universally superior replacement for cross-entropy.

1 Introduction

Fine-grained sentiment scales encode category and distance. In the five-level settings studied here, predicting *negative* for *very negative* is usually less severe than predicting *very positive*. We use fine-grained sentiment classification for ordered prediction beyond conventional positive/neutral/negative labels. All evaluated datasets instantiate this setting with five ordered sentiment labels. Yet a one-hot cross-entropy (CE) target gives both errors the same target geometry. This motivates objectives that redistribute target mass, decompose the scale into ordered decisions, or use learned thresholds; such objectives may reduce severe errors without improving exact match.

The intuitive appeal of ordinality is not evidence of reliable transfer to text. A pretrained language encoder already represents semantic regularities, while five-way sentiment labels can be noisy or ambiguous at adjacent boundaries. A stronger ordinal head can also reduce output flexibility: a cumulative link model (CLM), for example, compresses the encoder representation to one latent score before applying ordered thresholds. Whether that constraint is helpful depends on the task, the optimization protocol, and the error costs used for evaluation.

Prior NLP work has compared explicit ordinal losses with implicit label-semantic techniques and shown that improvements in ordinal metrics can coexist with weaker nominal performance (Kasa et al., 2024). Soft targets have also improved five-level customer-satisfaction prediction (Manderscheid and Lee, 2023). The `dlordinal` package collects several deep ordinal methods (Bérchez-Moreno et al., 2025), but its reported experiments concern image benchmarks. These studies establish the relevance of ordinal supervision across text and image settings, but they do not resolve whether QWK, MAE, and exact-match Accuracy identify the same leading objective under a shared sentiment-classification evaluation.

We ask three questions:

RQ1: Overall performance. Across datasets and configured training seeds, how do ordinal objectives compare with CE in mean performance and variability?

RQ2: Metric disagreement. Do QWK, MAE, and exact-match Accuracy rank methods differently, and do QWK gains coincide with changes in adjacent and severe errors?

RQ3: Conditions and failures. What dataset conditions and prediction signatures accompany success or failure, particularly for CLM-based heads?

We make three contributions. First, we provide a controlled comparison of eight existing objectives using common splits, encoders, and metrics. Second, building on prior evidence of ordinal–nominal trade-offs, we analyze metric disagreement as a metric-dependent evaluation and method-choice problem: under the observed aggregate test results, QWK, MAE, and exact-match Accuracy can identify different leading objectives, and QWK rankings need not track either exact matches or coarse tail-error rates. Third, we report negative results and bounded failure diagnostics for CLM-family heads, while separating observations from untested mechanisms. This is a benchmark and evaluation study, not a new loss or a claim of universal ordinal advantage.

The comparison spans SST-5, a balanced Yelp subset, and a financial composite constructed from records attributed to SemEval-2017 Task 5 and FiQA-2018 Task 1. The main study uses BERT on all three datasets. A rerun with the finance-domain, sentiment-adapted `ProsusAI/finbert` checkpoint (Araci, 2019) on the financial split provides a bounded checkpoint-condition check rather than a fourth task or a general encoder comparison.

2 Related Work

2.1 Ordinal Classification

Ordinal classification assumes labels $0, \ldots, K-1$ have a meaningful order. Classical reductions transform the task into $K-1$ binary problems (Frank and Hall, 2001). Modern neural variants differ in their output parameterization. CORN estimates conditional probabilities and combines them by the chain rule (Shi et al., 2023), whereas CLMs learn a latent scalar and ordered cut-points (McCullagh, 1980; Vargas et al., 2020). These approaches alter the output space, whereas distributional targets preserve a K-class head and change only the supervision.

Soft-label objectives encode neighborhood structure by assigning more target mass to classes near the gold label. We evaluate Beta (Vargas et al., 2022), Binomial (Liu et al., 2020), Exponential (Vargas et al., 2023a), and Triangular (Vargas et al., 2023b) variants. We also evaluate a two-stage CLM configuration whose second stage optimizes a differentiable quadratic weighted-kappa objective (de la Torre et al., 2018). Together with CE, CORN, and the standard CLM, these objectives define the eight-method comparison used throughout the paper.

2.2 Ordinal Text and Sentiment Classification

Kasa et al. (2024) compare explicit ordinal losses with implicit label-semantic approaches, including entailment-style and generative formulations. They propose a hybrid loss and evaluate weighted F1 together with MAE, MSE, and off-by-one accuracy on SNLI, Amazon Reviews, and SST-5, including reduced-data settings. Their results already show that improvements in ordinal metrics can coexist with weaker nominal performance. Our study builds on, rather than rediscovers, this trade-off. We focus on eight existing objectives—CE, four distributional soft-label objectives, CORN, CLM, and CLMWK—under shared dataset splits and evaluation metrics within each encoder–dataset block. The central question is whether QWK, MAE, and exact-match Accuracy lead to different model choices, and how those disagreements relate to QWK's marginal normalization, error distance, and prediction distributions (Cohen, 1968).

The image experiments reported with `dlordinal` show that CLMWK is competitive, and strongest among the package methods on some benchmarks (Bérchez-Moreno et al., 2025). They involve different inputs, encoders, and optimization regimes. Our comparison tests whether these objective families transfer to text; it cannot identify modality as the cause of any difference, nor does it compare every package method.

2.3 Evaluation Metrics

Exact-match Accuracy treats every wrong class equally. MAE distinguishes errors by absolute label distance. QWK uses quadratic distance weights and normalizes observed disagreement by disagreement expected from the true and predicted marginals (Cohen, 1968). Consequently, QWK can reward reducing a small number of severe errors while Accuracy falls, and two models with similar error distances can differ when their prediction distributions differ. Reporting only one metric can therefore change which objective appears

preferable. We make this disagreement an object of study rather than a reporting afterthought.

3 Methods

3.1 Task and Prediction Geometry

For an input text x, the task is to predict $y \in \{0, \ldots, K-1\}$, with $K = 5$ from most negative to most positive. Let $e = y - \hat{y}$. Exact predictions have $|e| = 0$; adjacent errors have $|e| = 1$; we separately report $|e| = 2$ and $|e| \geq 3$. This decomposition is useful because Accuracy depends only on the first category, MAE increases linearly with distance, and QWK uses squared distance inside a marginally normalized agreement statistic.

Table 1 summarizes the output and decision rule for every evaluated family. CE and all soft-label objectives use the same BERT pooled representation, dropout, and five-way linear head. CORN changes the head to four conditional logits. CLM and CLMWK reduce the pooled representation to a scalar and map it to class probabilities through ordered thresholds. Unless noted otherwise, the predicted label is the class with maximum reconstructed probability.

3.2 CE and Distributional Targets

Let $\mathbf{z} \in \mathbb{R}^K$ and $p(k \mid x) = \mathrm{softmax}(\mathbf{z})_k$. CE minimizes

$$\mathcal{L}_{\mathrm{CE}}(x, y) = -\log p(y \mid x). \tag{1}$$

The four soft-label methods replace the one-hot target with a normalized distribution $q_y(k; \eta)$ centered on the gold class:

$$\mathcal{L}_{\mathrm{soft}}(x, y) = -\sum_{k=0}^{K-1} q_y(k; \eta) \log p(k \mid x). \tag{2}$$

Beta, Binomial, Exponential, and Triangular targets differ in shape and tail mass; η controls concentration in the implementation (Bérchez-Moreno et al., 2025). The loss makes confusing nearby classes less costly during training, but inference remains ordinary five-way argmax. This isolates the effect of target geometry from a change in head capacity.

3.3 Conditional and Threshold Objectives

CORN outputs conditional logits for $P(y > j \mid y > j - 1, x)$. Writing their sigmoid values as p_j, the code reconstructs

$$P(y = 0 \mid x) = 1 - p_0, \tag{3}$$

$$P(y = k \mid x) = \left(\prod_{j=0}^{k-1} p_j \right) (1 - p_k), \quad 1 \leq k \leq K-2, \tag{4}$$

$$P(y = K-1 \mid x) = \prod_{j=0}^{K-2} p_j. \tag{5}$$

Each conditional loss is trained only on examples that reach its decision, following CORN (Shi et al., 2023).

CLM maps the pooled encoder vector to a scalar $s(x)$ and learns ordered thresholds $\tau_1 < \cdots < \tau_{K-1}$. With a logistic link,

$$P(y \leq k \mid x) = \sigma(\tau_{k+1} - s(x)), \quad 0 \leq k \leq K-2, \tag{6}$$

and class probabilities are differences of successive cumulative probabilities (Vargas et al., 2020). CLM minimizes their negative log-likelihood. CLMWK first fits the same likelihood and then fine-tunes with the differentiable weighted-kappa loss. The scalar bottleneck and threshold ordering impose more structure than the CE/soft-label head; this may regularize the task or restrict representations that are not well ordered along one axis.

4 Experimental Setup

4.1 Datasets and Splits

We use the five-class fine-grained task of the Stanford Sentiment Treebank (hereafter SST-5), which contains movie-review sentences labelled from very negative to very positive (Socher et al., 2013). The local CSV corresponds to the concatenated train and test partitions used by the `SetFit/sst5` loader and does not retain an original split identifier or a separate development partition. Because the shared training protocol requires development data for QWK-based hyperparameter selection and early stopping, we apply the same deterministic stratified 70/15/15 splitting procedure used for the other datasets after duplicate-text grouping. Collapsing 12 duplicate rows leaves 10,742 examples. This is a custom within-study repartition and is not intended for direct comparison with results obtained using the standard SST-5 partitions.

Table 1: Alignment between each evaluated objective, output representation, target or loss, and prediction rule.

Family	Output	Training signal	Prediction
CE	K logits	One-hot negative log-likelihood	Argmax of K-way softmax
Soft labels	K logits	Cross-entropy with Beta, Binomial, Exponential, or Triangular target distribution	Argmax of K-way softmax
CORN	$K-1$ conditional logits	Conditional binary losses on eligible examples	Argmax after chain-rule reconstruction of K probabilities
CLM	One scalar plus $K-1$ ordered thresholds	Negative log-likelihood of threshold-derived class probabilities	Argmax of threshold-derived probabilities
CLMWK	Same as CLM	CLM likelihood pretraining, then differentiable QWK loss	Argmax of threshold-derived probabilities

Yelp Review Full contains five-star review labels (Zhang et al., 2015). To keep the eight-objective, 20-seed comparison computationally tractable, the loader reads a fixed 50,000-record candidate pool from the local CSV, removes duplicate text groups, and samples 4,000 examples per class with random state 42. This produces a class-balanced subset of 20,000 reviews. The equal class counts control label prevalence within the study, but the subset is not a uniform sample of the complete Yelp corpus.

Financial composite. The financial composite combines the released training portions of SemEval-2017 Task 5 (Cortis et al., 2017) and FiQA-2018 Task 1 (Maia et al., 2018). It contains 1,710 SemEval Subtask 1 microblog records, 1,156 SemEval Subtask 2 financial news statements and headlines, and 1,111 FiQA Task 1 records, comprising 675 posts and 436 headlines. The heterogeneous source records were mapped to a common schema. For SemEval Subtask 1, the `text` field in the authors' internally archived input contains an annotated sentiment-bearing span; for SemEval Subtask 2 and FiQA, it contains the corresponding news statement, headline, or post sentence. The continuous sentiment annotation is retained in `score`; `source` identifies the medium as `post` or `headline`; and `dataset` identifies the originating task and subtask. This internally archived `new_code/merged_all.csv` contains 3,977 records and is not included in the accompanying anonymous software artifact.

Both source tasks provide continuous sentiment scores between -1 and 1. During construction of the archived composite, these scores were discretized into five ordered labels using $[-1.0, -0.5] \mapsto -2$, $(-0.5, -0.1] \mapsto -1$, $(-0.1, 0.1) \mapsto 0$, $[0.1, 0.5) \mapsto 1$, and $[0.5, 1.0] \mapsto 2$. The archived score–label pairs satisfy these boundaries for every record. The preparation script removes 20 SemEval Subtask 1 records whose `text` field is empty, yielding 3,957 records: 1,690 from SemEval Subtask 1, 1,156 from Subtask 2, and 1,111 from FiQA. No other record is removed, reordered, balanced, or deduplicated at this preparation stage. Because the training objectives require zero-based class indices, `new_code/prepare_merged_5class.py` subsequently applies

$$y_{\text{model}} = y_{\text{binned}} + 2, \tag{7}$$

mapping $-2, \ldots, 2$ to $0, \ldots, 4$. This shift changes only the numerical encoding and preserves the five class boundaries, order, and sentiment meanings. The script also adds a human-readable `label_name` field while preserving the surviving `text`, `score`, `source`, and `dataset` fields.

Before splitting, the runner converts text to string, strips leading/trailing whitespace, and lowercases it; internal whitespace is not collapsed. Duplicate rows sharing this exact key are merged; a label conflict is resolved by its unique majority label, while a group tied for highest frequency is dropped. A stratified 70/15/15 split then uses random state 42. For finance, the 3,957 model-ready rows form 2,961 text groups (996 rows relative to the unique-group count); 84 groups conflict, of which 45 have a retained majority and 39 ties are dropped, leaving 2,922 split examples. Table 2 gives the realized totals and splits.

The model-ready financial CSV is not a deduplicated dataset; grouping, conflict handling, and splitting are later runner operations. The authors' local inputs and deterministic code exactly reconstruct the SST-5 and Financial test manifests and confirm zero grouping-key overlap across their train/development/test splits. The

Table 2: Post-grouping dataset scale, realized splits, and label counts. Labels run from 0 (most negative) to 4 (most positive).

Dataset	Total	Train	Dev	Test	0	1	2	3	4
SST-5	10,742	7,518	1,612	1,612	1,369	2,847	2,012	2,828	1,686
Yelp	20,000	14,000	3,000	3,000	4,000	4,000	4,000	4,000	4,000
Financial	2,922	2,044	439	439	281	732	246	1,287	376

internally archived split JSON files store split sizes plus test texts and labels, not complete train/development/test indices. Because the local Yelp CSV is not packaged, its train/development partitions cannot be independently reconstructed here, although the internally archived test manifest aligns exactly with every corresponding prediction file.

4.2 Encoder, Preprocessing, and Controls

The main encoder is `bert-base-uncased` (Devlin et al., 2019). The financial condition uses the finance-domain, sentiment-adapted `ProsusAI/finbert` checkpoint (Araci, 2019).[1] This checkpoint incorporates finance-domain adaptation and subsequent sentiment-task fine-tuning on Financial PhraseBank. We load its BERT encoder with `AutoModel` and initialize a new task-specific classification or ordinal head for each objective. Relative to `bert-base-uncased`, this checkpoint-condition comparison changes both domain and prior task adaptation. It therefore does not isolate the effect of domain pretraining. Tokenization truncates/pads to 256 tokens for SST-5 and Yelp and 128 for Financial. CE, soft-label, and CORN models apply dropout 0.1 to the pooled output before their linear heads; CLM maps it directly to a scalar. Methods within an encoder–dataset block use the same split. CE-versus-soft-label comparisons hold the head fixed, whereas CORN and CLM change it.

Batch size is 128 and the optimizer is AdamW (Loshchilov and Hutter, 2019). CE, soft-label, and CORN runs use at most 15 epochs with early-stopping patience 3; CLM uses at most 50 with patience 5. CLMWK first fits likelihood for at most 50 epochs with patience 5, then fits weighted kappa for at most 10 epochs with patience 3 at one tenth of the selected Stage-1 learning rate. Candidate runs use at most 10 epochs with patience 3. Across candidate and final fits, including both CLMWK stages, early stopping monitors development loss and restores the best-loss checkpoint; the final candidate configuration is selected by development QWK. Soft-label tuning jointly searches learning rate $\{1, 2, 3, 5\} \times 10^{-5}$ and $\eta \in \{.3, .5, .7, .9\}$. These unequal schedules make runtime comparisons non-equivalent; selected values are in Appendix A.

Final results are reported as mean±SD over 20 predetermined seeds. All comparisons are descriptive; "highest," "lowest," and "outperforms" refer only to observed aggregate means and do not imply statistical significance. Archived tuning records retain candidate validation QWK values and selections, but the code does not explicitly reset the RNG before every candidate. We therefore do not interpret candidate comparisons as matched stochastic trials or infer what MAE- or Accuracy-directed tuning would select.

Summing the archived mean runtime of all 32 configurations and multiplying by 20 seeds gives a 26.8 accelerator-hour lower bound for final runs. It excludes tuning, preliminary or aborted attempts, and extra analyses. Archived environment records identify two NVIDIA GeForce RTX 4090 D GPUs (about 23.5GB each), CUDA 12.4, and PyTorch 2.6.0+cu124; the remaining environment was not frozen. The seed list is in Appendix A.

4.3 Metrics and Analyses

Let n be the test-set size. For test example i, y_i and $\hat{y}_i$ are the gold and predicted labels, and $\mathbf{1}\{\cdot\}$ is the indicator function. Accuracy is $n^{-1}\sum_{i=1}^{n}\mathbf{1}\{y_i = \hat{y}_i\}$, and MAE is $n^{-1}\sum_{i=1}^{n}|y_i - \hat{y}_i|$. For QWK, $a, b \in \{0, \ldots, K-1\}$ index gold-label rows and predicted-label columns. O_{ab} is the observed number of examples with gold a and prediction b; E_{ab} is the corresponding expected count under the product of the two marginals. With $w_{ab} = (a-b)^2/(K-1)^2$,

$$\mathrm{QWK} = 1 - \frac{\sum_{a,b} w_{ab} O_{ab}}{\sum_{a,b} w_{ab} E_{ab}}. \qquad (8)$$

[1] https://huggingface.co/ProsusAI/finbert

Thus, a indexes the true-label row and b indexes the predicted-label column in both O and E. Higher Accuracy and QWK and lower MAE are better. The expected matrix makes QWK sensitive not only to error distance but also to predicted class prevalence.

We report mean±SD aggregates for RQ1. For RQ2, we compare metric-dependent leading objectives and CE-relative directions across the four soft-label objectives. We additionally recompute QWK and error-distance profiles from the same seed-42 prediction files, ensuring that each diagnostic comparison is made at the level of one captured run. These single-run diagnostics are descriptive and are not used to estimate uncertainty.

5 Results

5.1 RQ1: Overall Performance

Table 3 reports the BERT aggregates. On SST-5, Beta, Binomial, and Exponential tie at the highest archived three-decimal QWK (.769); Exponential has the lowest MAE (.567), and Triangular the highest Accuracy (.518). On Yelp, Binomial has the highest QWK (.865), CE/Binomial/Triangular tie for lowest MAE (.427), and Triangular has the highest Accuracy (.623). Financial results separate the metrics most clearly: Binomial raises observed QWK from CE's .665 to .684, but MAE worsens from .537 to .560 and Accuracy falls from .604 to .580.

Across 12 soft-label-versus-CE comparisons, mean QWK is higher in eight, tied in one, and lower in three. Accuracy is higher in only three, tied in one, and lower in eight; MAE is lower in two, tied in three, and higher in seven. Ties and directions use the archived three-decimal means. Thus, soft labels more often raise observed QWK, but those gains frequently do not transfer to the other metrics. These are descriptive aggregate differences, not significance-tested effects.

5.2 RQ2: Metric Disagreement

The metric-dependent leader summary in Table 4 makes the evaluation disagreement explicit. Under the observed aggregate test results from the same trained models, SST-5 and Yelp have different leading objectives across metrics, while Financial has Binomial leading by QWK and CE leading by both MAE and Accuracy. Only ties at the archived three-decimal precision are shown.

Financial Binomial supplies a direct QWK–MAE–Accuracy reversal: relative to CE, its observed QWK is $+.019$, MAE is $+.023$ (worse), and Accuracy is $-.024$. A higher QWK therefore does not imply either more exact predictions or lower absolute error. The metrics weight squared ordinal disagreement with marginal normalization, absolute distance, and exact matching differently; disagreement is not evidence that any one metric is intrinsically wrong.

The seed-42 predictions allow a same-run comparison of QWK and error distance. Binomial exceeds CE in QWK on SST-5 (.780 versus .758) and Financial (.692 versus .665), yet also has larger shares of errors spanning two or more classes (8.37% versus 6.95% and 11.62% versus 11.39%). On Yelp, QWK is tied to three decimals (.864), but those shares are 3.50% versus 3.43%. Thus, QWK and the coarse tail-error rate need not move together within a captured run.

QWK also depends on the true and predicted label marginals through E_{ab}. Reducing distant errors can improve its numerator, while changing predicted prevalence changes its normalization. Candidate QWK values are logged, but all hyperparameters were selected by validation QWK; we did not run MAE- or Accuracy-directed counterfactual tuning. RQ2 therefore demonstrates metric-dependent *evaluation*, not metric-specific model selection.

5.3 RQ3: Conditions and CLM-Family Head Failures

Both CLM-family heads have lower observed QWK than CE on all three BERT datasets. CLMWK reaches .683/.776/.628 on SST-5/Yelp/Financial versus CE's .759/.863/.665, and it also has worse MAE and Accuracy. This pattern persists in the FinBERT condition (.655 versus .712 QWK), but it remains specific to these heads and schedules.

Figure 2 provides a more specific seed-42 diagnostic. CLMWK assigns 88.83%, 94.43%, and 98.41% of SST-5, Yelp, and Financial predictions to classes 0 or 4. Both CLM and CLMWK assign no prediction to intermediate classes 1 or 3 on any dataset. This coincides with their lower aggregate Accuracy and higher MAE, but it is descriptive single-run evidence, not proof that the scalar bottleneck, thresholds, or kappa stage independently causes the failure.

Balance alone cannot explain the structured-

Table 3: BERT test results as mean ± SD over 20 predetermined seeds. Bold marks the highest archived three-decimal mean per dataset/metric (lowest for MAE), including ties; it does not denote statistical significance.

Dataset	Method	QWK↑	MAE↓	Accuracy↑
SST-5	CE	.759 ±.010	.573 ±.012	.514 ±.008
	Beta	**.769 ±.007**	.590 ±.015	.501 ±.010
	Binomial	**.769 ±.008**	.573 ±.015	.514 ±.011
	Exponential	**.769 ±.010**	**.567 ±.014**	.517 ±.011
	Triangular	.768 ±.008	.572 ±.016	**.518 ±.011**
	CORN	.766 ±.009	.569 ±.008	.511 ±.007
	CLM	.697 ±.012	.926 ±.024	.316 ±.007
	CLMWK	.683 ±.008	.985 ±.021	.299 ±.007
Yelp	CE	.863 ±.002	**.427 ±.009**	.622 ±.005
	Beta	.862 ±.004	.441 ±.010	.614 ±.006
	Binomial	**.865 ±.004**	**.427 ±.010**	.621 ±.006
	Exponential	.864 ±.003	.434 ±.008	.617 ±.005
	Triangular	.863 ±.002	**.427 ±.009**	**.623 ±.005**
	CORN	.864 ±.005	.431 ±.014	.618 ±.008
	CLM	.792 ±.010	.717 ±.028	.461 ±.011
	CLMWK	.776 ±.012	.776 ±.035	.436 ±.014
Financial	CE	.665 ±.029	**.537 ±.026**	**.604 ±.014**
	Beta	.668 ±.037	.554 ±.035	.589 ±.016
	Binomial	**.684 ±.033**	.560 ±.037	.580 ±.021
	Exponential	.659 ±.048	.551 ±.035	.594 ±.018
	Triangular	.648 ±.034	.563 ±.032	.589 ±.016
	CORN	.664 ±.036	.545 ±.030	.597 ±.014
	CLM	.650 ±.023	1.017 ±.035	.222 ±.008
	CLMWK	.628 ±.017	1.092 ±.022	.213 ±.004

Table 4: Metric-specific choices by observed aggregate mean.

Dataset	QWK	MAE	Acc.
SST-5	Beta/Bin./Exp.	Exponential	Triangular
Yelp	Binomial	CE/Bin./Tri.	Triangular
Financial	Binomial	CE	CE

head pattern: it occurs on the balanced Yelp subset and the skewed SST-5 and Financial datasets. Nor does the FinBERT checkpoint remove it. The packaged datasets do not contain repeated ratings or annotation distributions, however, so adjacent-label ambiguity remains an untested explanation rather than a measured property.

The FinBERT condition further illustrates configuration dependence. On Financial, Exponential has the best observed mean on all three metrics under the FinBERT checkpoint: .718±.025 QWK, .500±.020 MAE, and .613±.013 Accuracy, versus CE's .712±.025/.508±.022/.610±.012. All eight objectives have higher observed QWK under the FinBERT checkpoint than under BERT in this financial condition (CE +.047; Exponential +.059). Because the checkpoint differs in both domain and prior sentiment-task adaptation, this comparison cannot identify which component accounts for the observed differences. It is one dataset-specific checkpoint-condition comparison, not evidence that FinBERT is generally superior. Full results are in Appendix B.

The safest transfer claim is configuration-specific. CLMWK is reported as the strongest evaluated package method on some image benchmarks (Bérchez-Moreno et al., 2025); our BERT setup changes modality, encoder geometry, head capacity, and schedule. The present results show that a scalar ordered head and two-stage kappa training did not transfer reliably in these configurations. Isolating why would require controlled ablations of the scalar bottleneck, threshold parameterization, dropout, learning rate, and two-stage schedule.

6 Discussion and Practical Guidance

The results favor a cost-sensitive view of method choice. If exact rating prediction matters most, an Accuracy-leading model may be preferable even when its QWK is lower. If ordinal distance matters, QWK or MAE may be preferable, but their rankings can also disagree. Each should be accompanied by Accuracy, an error-distance histogram, and predicted-label marginals.

Among the tested approaches, soft labels are the lowest-friction ordinal intervention because they preserve the five-way head. Their QWK directions are more often favorable than not across the three BERT datasets, but gains less often carry to MAE or Accuracy. Financial Binomial exhibits a three-metric conflict under BERT, whereas Exponential has the highest observed means for all three metrics in the FinBERT condition. Metric alignment therefore also depends on the checkpoint and broader configuration. The CLM-family heads require diagnostics and ablations before adoption.

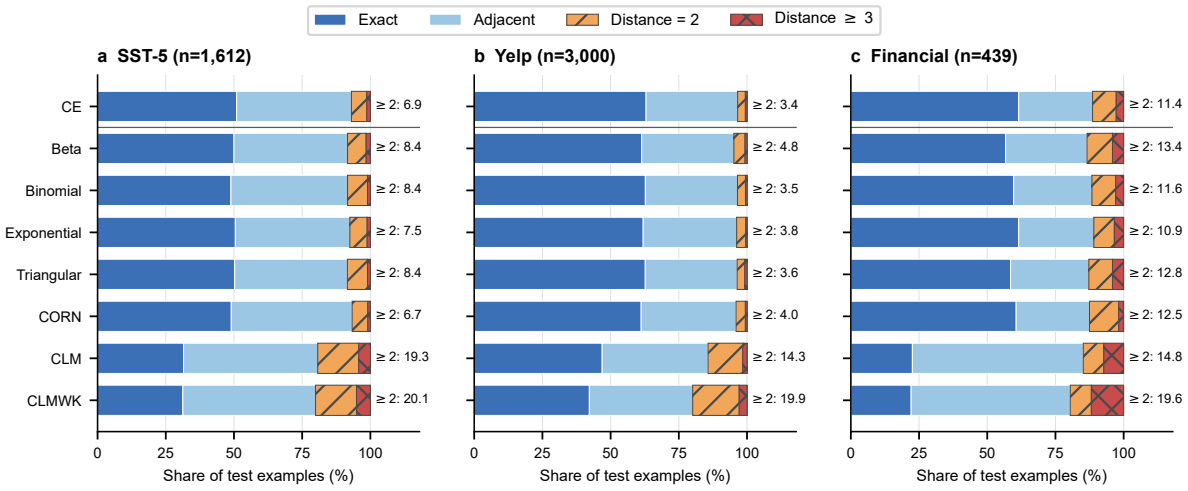


Figure 1: Seed-42 composition of exact predictions, adjacent errors, errors spanning exactly two classes, and errors spanning three or more classes. Right-side labels give the share spanning two or more classes. Each method's predictions follow its formal training protocol and provide descriptive single-run evidence, not an uncertainty estimate.

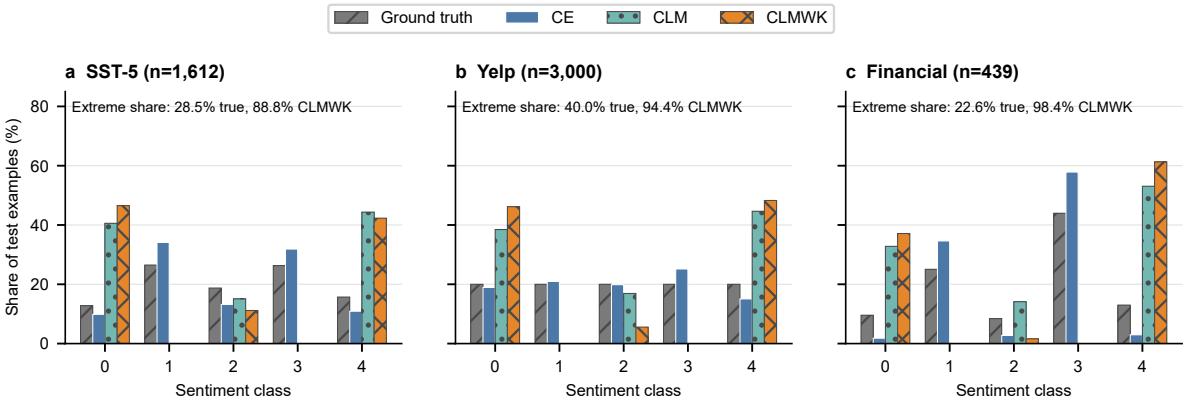


Figure 2: Seed-42 true and predicted label distributions for BERT. CLM and CLMWK leave classes 1 and 3 unused and concentrate predictions at the extremes. This is descriptive single-run evidence, not an uncertainty estimate.

The study also illustrates a reporting risk. Selecting hyperparameters by QWK and presenting only QWK can hide an Accuracy decline; presenting only Accuracy can hide a reduction in severe errors. We recommend declaring error costs before selection, using the primary metric that matches those costs, and retaining at least one complementary exact and one distance-sensitive measure. Where class prevalence matters, prediction distributions should be reported because QWK's expected-agreement term depends on them.

These recommendations are bounded by descriptive aggregate comparisons on three English fine-grained sentiment datasets with five ordered labels, one main encoder, and one financial checkpoint. They establish neither paired effects, a universal best objective, nor causes of the structured-head failures.

7 Conclusion

This controlled study finds no universal ordinal winner here. Across three BERT datasets, soft labels more often raise observed QWK than CE, but not consistently MAE or Accuracy. Seed-42 analyses show that QWK need not track the share of errors spanning two or more classes and expose extreme CLM-family predictions. The finance-domain, sentiment-adapted checkpoint changes the financial ordering. Objective choice therefore depends on error costs, checkpoint, and prediction behavior.

Limitations

Comparisons use aggregate mean±SD without paired inference. Candidate QWK values are retained, but candidate-wise RNG state was not explicitly reset, and MAE- or Accuracy-directed counterfactual selection was not run. The seed-42 analyses are descriptive single-run diagnostics, not uncertainty estimates.

The local Yelp CSV and full train/development indices are not packaged: only the loaded-candidate process, split sizes, test texts, and test labels are internally archived, so the sampled train/development partitions cannot be independently rebuilt here. The internally archived split JSON files likewise do not retain full indices.

Financial provenance boundary. The official task sources, applicable usage conditions, common field schema, label discretization, and repository-internal transformations are documented in Appendix F. The authors retain an internal archived copy of `merged_all.csv` from which the model-ready financial data and all subsequent grouping and splitting operations can be reproduced deterministically. The accompanying anonymous software artifact provides the preparation and training code, aggregate result summaries, field mappings, processing statistics, and reconstruction instructions. Restricted source text and text-bearing split or prediction derivatives are excluded under the source-data terms. The original one-off script that selected fields from the official downloads and constructed `merged_all.csv` was not preserved, so the artifact cannot reconstruct that archived composite byte for byte from the official downloads. This is an upstream provenance and external reproducibility boundary rather than an uncertainty in the internally archived sample counts, labels, experimental splits, or reported results.

Our empirical scope comprises three English fine-grained sentiment datasets, each represented with five ordered labels, BERT-base-scale encoders, and eight evaluated objectives: CE, four soft-label objectives, CORN, CLM, and CLMWK. Larger encoders, multilingual data, and additional tasks are outside this scope. The `ProsusAI/finbert` condition jointly changes finance-domain adaptation and prior sentiment-task fine-tuning, so it does not isolate the effect of domain pretraining. Label-boundary ambiguity is not directly measured. CLM/CLMWK structure and schedules were not ablated, so their failures cannot be attributed causally. The 26.8 accelerator-hour figure excludes tuning, aborted attempts, and extra analyses, and the server's complete dependency freeze is unavailable.

Ethical Considerations

The work makes no deployment or safety claim. Sentiment labels compress context and may encode annotator or domain biases. Financial sentiment outputs could be misused as trading advice and should not be presented as decision-ready signals. Source text may contain offensive language, personal references, or platform-specific content. The accompanying anonymous software artifact excludes restricted source text and text-bearing derivatives under the source-data terms. The manuscript and analysis were reorganized with generative-AI assistance; the authors remain responsible for source verification, code, results, and disclosure under the venue policy.

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A Reproducibility Details

Seeds and schedules. The predetermined seeds are 42, 123, 1024, 2026, 3407, 4096, 5150, 6024, 7777, 8888, 101, 202, 303, 404, 505, 606, 707, 808, 909, and 1111. Final CE, soft-label, and CORN runs use at most 15 epochs with patience 3; CLM uses at most 50 with patience 5. CLMWK uses at most 50 Stage-1 likelihood epochs with patience 5 and then at most 10 Stage-2 weighted-kappa epochs with patience 3; its second-stage learning rate is 0.1 times the selected first-stage rate. Candidate runs use at most 10 epochs with patience 3. Across candidate and final fits, including both CLMWK stages, early stopping monitors validation loss and restores the best-loss checkpoint; the final candidate configuration is selected by validation QWK. Batch size is 128. Maximum token length is 256 for SST-5/Yelp and 128 for Financial. Seed-42 predictions are captured within the formal 20-seed loop and are used only for descriptive error analysis.

Table 5: Selected BERT learning rate (LR) and soft-target concentration η. Dashes denote objectives without an η search.

Method	SST-5 LR	SST-5 η	Yelp LR	Yelp η	Financial LR	Financial η
CE	2e−5	–	2e−5	–	5e−5	–
Beta	1e−5	.9	1e−5	.9	2e−5	.9
Binomial	5e−5	.5	3e−5	.5	3e−5	.7
Exponential	3e−5	.7	1e−5	.9	3e−5	.5
Triangular	3e−5	.5	2e−5	.5	3e−5	.3
CORN	1e−5	–	2e−5	–	5e−5	–
CLM	5e−5	–	5e−5	–	5e−5	–
CLMWK	1e−4	–	1e−4	–	5e−5	–

Table 6: Selected FinBERT hyperparameters on Financial.

Method	LR	η
CE	5e−5	–
Beta	3e−5	.5
Binomial	5e−5	.3
Exponential	3e−5	.5
Triangular	2e−5	.9
CORN	2e−5	–
CLM	5e−5	–
CLMWK	1e−4	–

Software, hardware, and compute. The final-run logs record two NVIDIA GeForce RTX 4090 D GPUs with about 23.5GB each, CUDA 12.4, and PyTorch 2.6.0+cu124. The repository lock file describes a different environment and is not treated as a freeze of this server run; other exact dependency versions were not archived. The study uses PyTorch (Paszke et al., 2019) and Transformers (Wolf et al., 2020). Summing mean runtime for the 32 configurations and multiplying by 20 gives a 26.8 accelerator-hour lower bound, excluding tuning, preliminary or aborted attempts, and additional analyses.

Seed-42 decomposition. Table 7 reports the complete BERT diagnostic used by the figures and main-text severe-error comparison.

Table 7: BERT seed-42 outcomes as percentages of each test set (SST-5 $n = 1{,}612$; Yelp $n = 3{,}000$; Financial $n = 439$). Categories are mutually exclusive and sum to 100%. These single-run diagnostics are not uncertainty estimates.

Dataset	Method	Exact	Adjacent	Distance $= 2$	Distance ≥ 3
SST-5	CE	50.99	42.06	5.65	1.30
	Beta	50.00	41.63	6.82	1.55
	Binomial	48.88	42.74	7.32	1.05
	Exponential	50.50	42.00	6.33	1.18
	Triangular	50.25	41.32	7.38	1.05
	CORN	49.01	44.29	5.77	0.93
	CLM	31.51	49.19	15.01	4.28
	CLMWK	31.27	48.64	15.01	5.09
Yelp	CE	63.00	33.57	2.83	0.60
	Beta	61.37	33.83	4.07	0.73
	Binomial	62.73	33.77	2.93	0.57
	Exponential	61.93	34.27	3.33	0.47
	Triangular	62.70	33.67	2.83	0.80
	CORN	61.20	34.77	3.43	0.60
	CLM	46.90	38.80	12.73	1.57
	CLMWK	42.20	37.90	17.03	2.87
Financial	CE	61.50	27.11	8.66	2.73
	Beta	56.72	29.84	9.34	4.10
	Binomial	59.68	28.70	8.66	2.96
	Exponential	61.50	27.56	7.52	3.42
	Triangular	58.54	28.70	8.66	4.10
	CORN	60.59	26.88	10.71	1.82
	CLM	22.55	62.64	7.52	7.29
	CLMWK	22.10	58.31	7.74	11.85

B FinBERT Financial Condition

Table 8 reports the FinBERT financial condition under the same aggregate reporting protocol.

Table 8: FinBERT Financial test results as mean ± SD over 20 predetermined seeds. Bold follows the archived three-decimal observed means and does not denote statistical significance.

Method	QWK↑	MAE↓	Accuracy↑
CE	.712 ±.025	.508 ±.022	.610 ±.012
Beta	.715 ±.024	.506 ±.022	.610 ±.017
Binomial	.707 ±.033	.517 ±.027	.604 ±.015
Exponential	**.718 ±.025**	**.500 ±.020**	**.613 ±.013**
Triangular	.710 ±.020	.512 ±.017	.607 ±.012
CORN	.703 ±.017	.509 ±.020	.608 ±.012
CLM	.675 ±.017	.983 ±.025	.230 ±.006
CLMWK	.655 ±.023	1.054 ±.030	.215 ±.004

C Financial Data, Split, and Rerun Audit

The public academic sources are SemEval-2017 Task 5 (Cortis et al., 2017) and FiQA-2018 Task 1 (Maia et al., 2018). The authors' internally archived input, `new_code/merged_all.csv`, contains 3,977 rows: 1,710 from SemEval Subtask 1, 1,156 from Subtask 2, and 1,111 from FiQA. It is not part of the accompanying anonymous software artifact. Removing 20 missing-text rows from Subtask 1 yields 3,957 rows (1,690/1,156/1,111). No other row is removed, reordered, balanced, or deduplicated. The surviving `text`, `score`, `source`, and `dataset` fields are unchanged.

Source labels $-2, \ldots, 2$ are shifted by two to model indices $0, \ldots, 4$; the score boundaries, ordinal categories, and meanings are unchanged. The preparation also adds `label_name`. Running `prepare_merged_5class.py` reproduces the internally archived model-ready file byte for byte in the authors' verification.

The runner's lowercase-plus-strip key turns 3,957 rows into 2,961 text groups, leaving 996 duplicate rows relative to the group count. Of 84 conflicting-label groups, 45 have a unique majority label and are retained; 39 highest-frequency ties are dropped. The resulting 2,922 examples have label counts 281/732/246/1,287/376 and split into 2,044/439/439 train/development/test examples. Deterministic reconstruction within the authors' internal archive matches the archived test texts and labels exactly and finds zero grouping-key overlap across splits. The internal output identifier `FiQA` denotes this full composite, not a FiQA-only dataset.

The formal experiment package contains 32 unique aggregate configurations: BERT on three datasets with eight methods and FinBERT on Financial with eight methods. It has no missing metric cells or duplicate keys. The 32 internally archived text-bearing prediction files contain 43,920 rows in total, agree with the internally combined prediction file as a row multiset, and align exactly with their split test arrays. These prediction and split files are internal verification evidence rather than contents of the accompanying anonymous software artifact. Completion records show that the formal run finished without a final-run failure.

The internally archived split JSON files contain split sizes, test texts, and test labels, but not complete train/development/test indices. The available code also does not reconstruct `merged_all.csv` byte for byte from the official SemEval/FiQA downloads; the original field-selection and merge script remains the upstream provenance boundary.

D Data and Evidence Audit

Runner-compatible reconstruction reduces SST-5 from 10,754 rows to 10,742 grouping keys by collapsing 12 duplicate rows. It finds no conflicting-label groups, exactly reproduces the 7,518/1,612/1,612 split and test manifest, and finds zero grouping-key overlap across splits.

For Yelp, logs record 50,000 loaded candidate rows and 49,995 candidate text groups before balanced sampling to 20,000 examples. The 14,000/3,000/3,000 split sizes and test array are archived and predictions align with that test array. Because the local raw CSV and full indices are not packaged, the exact sampled train/development partitions cannot be independently reconstructed from this repository.

All manuscript result rows, winner ties, error-distance summaries, prediction marginals, hyperparameters, and the runtime lower bound were regenerated from the archived formal-run outputs.

E Reproducibility Boundaries

The authors' internal evidence package supports the reported aggregate results, archived predictions, test-set alignment, and documented downstream data reconstruction. The accompanying anonymous software artifact provides the prepara-

tion and training code, aggregate result summaries, field mappings, processing statistics, and reconstruction instructions. Restricted source text and text-bearing split or prediction derivatives are excluded under the source-data terms. Alternative hyperparameter studies optimizing MAE or Accuracy are outside the evaluation protocol. Complete Yelp source-reconstruction metadata and train/development indices, a byte-identical reconstruction of `merged_all.csv` from the original task files, and a complete dependency freeze from the run server are not available in the accompanying anonymous software artifact.

Direct measurements of annotation uncertainty and matched CLM/CLMWK ablations are also outside the available evidence. Accordingly, the paper does not make causal claims about annotation ambiguity or about which component of the CLM-family configurations produced their observed behavior. These boundaries limit upstream reconstruction and causal interpretation where explicitly stated; they do not change the archived predictions or descriptive results reported in the paper.

F Data and Code Availability

Data access and licensing. SemEval-2017 Task 5 was distributed through the official task data page and requires acceptance of its academic data-release agreement. The agreement licenses the task annotations under CC BY-NC-SA 4.0 and specifies that Twitter and StockTwits content remains subject to the corresponding platform terms. FiQA-2018 Task 1 was distributed through the official FiQA challenge page, which designates its training and test data for non-commercial use. To respect these conditions, source texts should be obtained through the official task distributions rather than redistributed. The accompanying anonymous software artifact provides the preparation and training code, aggregate result summaries, field mappings, processing statistics, and reconstruction instructions. Restricted source text and text-bearing split or prediction derivatives are excluded under the source-data terms.

SST-5 and Yelp Review Full should likewise be obtained from their public distributions under their respective licenses.